TRINAY

AI-ASSISTED RETINAL SCREENING REPORT

CONFIDENTIAL MEDICAL DOCUMENT

DATE: 11 Oct 2026

DEPARTMENT OF TELE-OPHTHALMOLOGY & DIABETIC RETINOPATHY SCREENING

1. PATIENT INFORMATION

Patient Name:	Patient ID:	Age / Gender:	Patient Phone:
Aviraj abasaheb Murkute	**PX-001**	**20 Yrs / Male**	**9762481967**
Screening Date:	Examined Eye:	Screening Nurse:	Diabetes Duration:
11 Oct 2026	**Right Eye**	**avi**	**5 Years**

2. CLINICAL EXAMINATION

Visual Acuity: *Not available* Intraocular Pressure: *Not available* Existing Ocular Findings: **EX, HE, MA, SE**

VA / IOP have no fields in the current screening record and are not inferred.

3. IMAGE QUALITY ASSESSMENT

Image Filename: **0af338c60d2d.png** Image Quality: **Unsuitable** Quality Status: **Diagnostic Grade**

4. AI SCREENING RESULTS

Moderate Non-Proliferative DR (Level 2)

Deep neural ensemble prediction calibrated for primary tele-screening.

Macular edema assessment: Not available — no dedicated DME classifier output in the current pipeline.

CONFIDENCE SCORE

97.1%

Calibrated Model Output

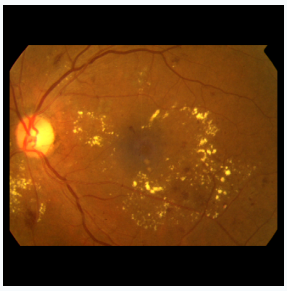
5. AI-DETECTED RETINAL FINDINGS

Microaneurysm	**Detected (642 px)**
Retinal Hemorrhage	**Detected (1,859 px)**
Hard Exudate	**Detected (12,277 px)**
Soft Exudate (Cotton-Wool Spot)	**Detected (48 px)**

5. RETINAL ANATOMY LOCALIZATION

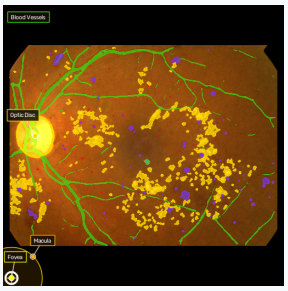
Optic Disc	**4,040 px (Localized)**
Macula / Fovea	**Pos: (0.25, 511)**
Retinal Blood Vessels	**10,938 px**
Media Clarity	**Unsuitable**

6. RETINAL VISUAL EVIDENCE GALLERY



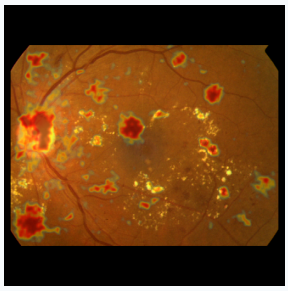
1. Original Fundus Image

True-colour capture; overlay reference



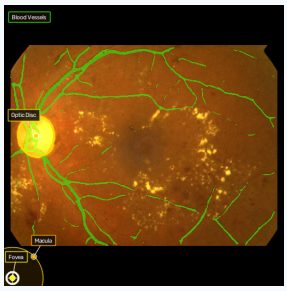
2. Combined Retinal AI Overlay

Landmarks & detected lesions



3. Grad-CAM++ Attention Map

Neural activation heatmap



4. Optic Disc and Retinal Vessels

Disc, macula & vessel landmarks

7. AI FINDINGS AND EXPLANATIONS

Microaneurysm — Detected: 642 px

What it is: A small localized outpouching of a retinal capillary wall where weakened pericytes allow dilation.

Why it matters: Microaneurysms represent the earliest observable retinal microvascular finding associated with diabetic retinopathy.

Clinical Note: The finding should be interpreted together with other retinal abnormalities and the patient's clinical glycemic history.

Retinal Hemorrhage — Detected: 1,859 px

What it is: Bleeding caused by leakage from damaged and ruptured retinal microvessels into intraretinal layers.

Why it matters: Retinal hemorrhages can occur in diabetic retinopathy and other retinal vascular disorders, indicating microvascular compromise.

Clinical Note: Requires clinical correlation and ophthalmologist assessment to evaluate the extent of vascular compromise.

Hard Exudate — Detected: 12,277 px

What it is: Lipid and lipoprotein deposits precipitated in the retina from abnormal vascular permeability and serum leakage.

Why it matters: They indicate chronic retinal vascular leakage and are important when assessing diabetic retinal disease and proximity to the fovea.

Clinical Note: Evaluate spatial relationship to the foveal avascular zone to monitor risk of diabetic macular edema (DME).

Soft Exudate (Cotton-Wool Spot) — Detected: 48 px

What it is: A localized area of retinal nerve-fiber-layer ischemic injury with axoplasmic flow stasis.

Why it matters: It represents focal retinal ischemic change and can occur in diabetic retinopathy as well as other systemic microvascular conditions.

Clinical Note: Reflects localized precapillary arteriolar occlusion and localized retinal ischemia rather than true exudation.

8. CLINICAL SUMMARY AND VALIDATION

4 retinal lesion types were detected in this screening, including Microaneurysm, Retinal Hemorrhage, Hard Exudate, Soft Exudate (Cotton-Wool Spot), encompassing a total quantified lesion area of 14,826 px. These findings represent retinal abnormalities that may be associated with diabetic retinal vascular disease. Validated neural classifier staging indicates Moderate Non-Proliferative DR (Level 2) (97.1% confidence). Pixel measurements reflect localized lesion load and must be correlated with clinical slit-lamp findings. In-person ophthalmologist review and dilated biomicroscopy are recommended.

Total detected lesion footprint: 14,826 px • Staging derived strictly from the calibrated ensemble, not lesion area alone. Shown exactly as returned; no reconciliation applied.

Retinal screening examination reveals findings consistent with Moderate Non-Proliferative DR (Level 2). Visible vascular anomalies detected in retinal vasculature. Recommendation: Vitreo-retinal specialist consultation for dilated biomicroscopy and optical coherence tomography (OCT) to rule out diabetic macular edema (DME).

What this means for the patient: The scan detected signs of diabetic eye changes that need attention from an eye specialist. Early detection allows gentle, effective treatments to protect your eyesight.

9. REFERRAL AND FOLLOW-UP

Referral Status:	Priority Level:	Target Facility:
Referred to Specialist	**Medium Priority**	**Regional Eye Care Center - Vitreo-Retina**

Referral Reason: **Diabetic retinopathy findings requiring clinical slit-lamp dilated examination.**

10. OPHTHALMOLOGIST REVIEW AND DISCLAIMER

Specialist Review: Pending

Specialist Review: **Pending**

Specialist Assessment: *Pending specialist clinical evaluation.*

Specialist Notes: *This screening record is queued for tele-ophthalmology verification.*

Specialist Recommendation: *Pending clinical review by assigned specialist.*

9. REFERRAL AND FOLLOW-UP — FOLLOW-UP (CONTINUED)

Comprehensive dilated ophthalmology evaluation within 2 to 4 weeks. Continue blood glucose and blood pressure monitoring.

10. OPHTHALMOLOGIST REVIEW AND DISCLAIMER — DISCLAIMER

TRINAY provides AI-assisted screening support and does not replace diagnosis by a qualified eye-care professional. Definitive clinical decisions should always be confirmed through direct dilated slit-lamp biomicroscopy or OCT by an ophthalmologist.